

REPLICATION & EXTENSION OF HODLER & RASCHKY (2014), QJE

Regional Favoritism

Nighttime Lights, Leader Birthplaces, and the Geography of Political Redistribution

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Applied Econometrics Using GIS Techniques (SoSe 2026)

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DMSP-OLS · 1994

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- Introduction
- Literature
- Data Sources
- Empirical Framework
- Extension
- Conclusion

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Introduction



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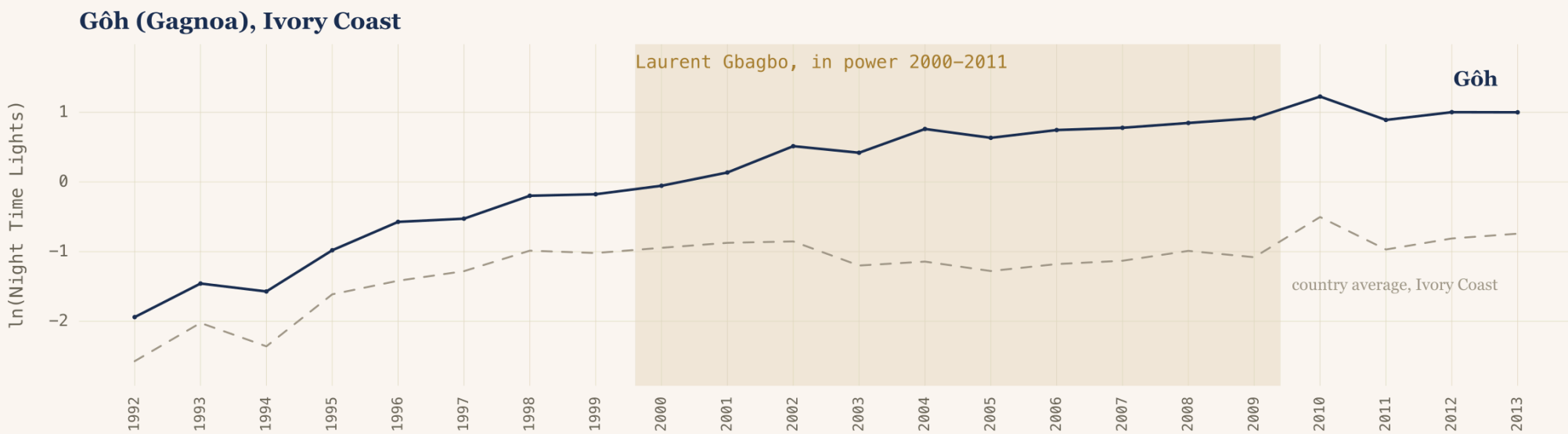
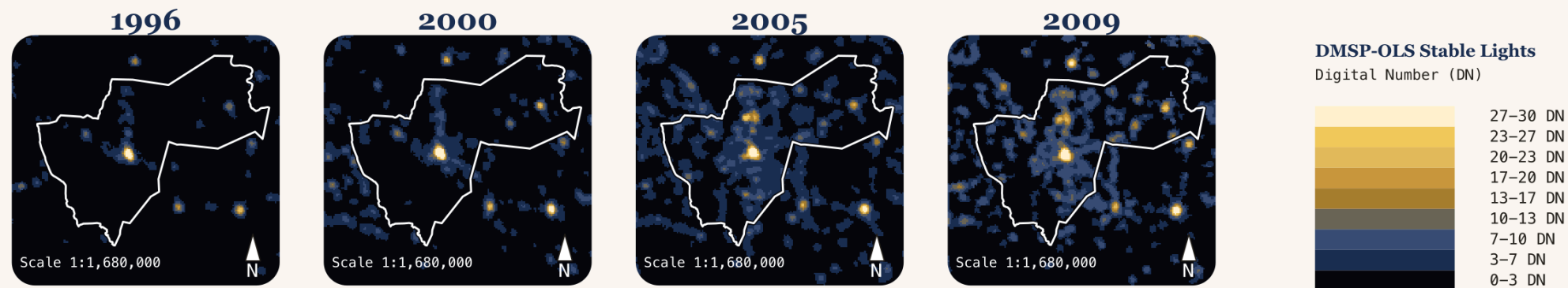
What's Regional Favoritism?

The tendency of national leaders to direct disproportionate public resources toward their own birth region, rather than allocating them on the basis of need or national interest.

How It's Measured

- Compare a leader's birth region against all other regions of the same country, in the same year
- Use satellite-recorded nighttime light intensity as a proxy for local economic activity
- Ask whether the birth region gets systematically brighter while its leader is in power

An Example: Ivory Coast



The Paper We Replicate

Hodler & Raschky (2014), *Quarterly Journal of Economics*

RESEARCH QUESTION

➤ Do national leaders steer resources to the region they were born in? And if so, how far does the effect reach geographically, how long does it last, and where is it worst?

SAMPLE

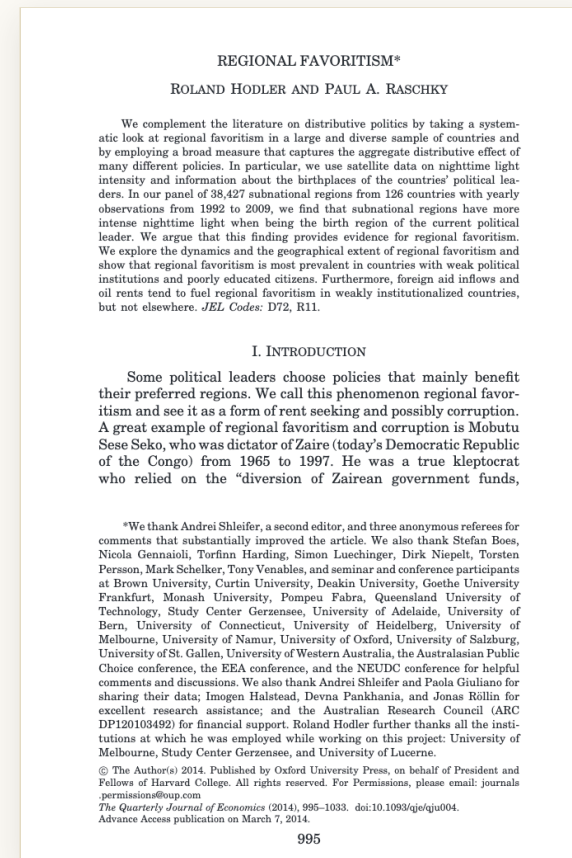
➤ **38,427 subnational regions in 126 countries**, observed annually from **1992 to 2009**, covering the entire usable DMSP-OLS satellite nighttime light record at the time.

EMPIRICAL STRATEGY

➤ The outcome is nighttime light intensity. A region is treated in the years its country's effective leader was born there. Region and country-year fixed effects hold the comparison **within country, within year**.

FINDING

➤ Birth regions are about **4% brighter** while their leader is in office. The effect is strongest where institutions are weak and education low, and it is amplified by foreign aid and oil rents ([Hodler & Raschky, 2014](#)).



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Literature



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FOUNDATIONAL

Measuring Economic Growth from Outer Space ([Henderson et al., 2012](#))

- **Data:** Nighttime lights matched to national GDP growth, cross-country panel
- **Scope:** Validating satellite light growth as an economic-activity proxy
- **Finding:** Light growth is an **independent proxy for GDP growth** that outperforms official statistics where national accounts are weak

Distributive Politics Around the World ([Golden & Min, 2013](#))

- **Data:** Review of the distributive-politics literature, no single dataset
- **Scope:** How political favoritism has been measured across countries
- **Finding:** Prior work relied almost entirely on **single-country spending data**, leaving systematic cross-country evidence largely absent

RELATED / EXTENSIONS

Does the Leader's Ethnicity Matter? ([Franck & Rainer, 2012](#))

- **Data:** Demographic and Health Surveys, Sub-Saharan Africa
- **Scope:** Ethnic favoritism in health and education outcomes
- **Finding:** Districts sharing the president's ethnicity see **relative gains in education and child mortality** while that leader is in power

Birthplace Favoritism Revisited ([Bora, 2025](#))

- **Data:** VIIRS nighttime lights, 2012-2025, matched to PLAD birthplaces and GADM boundaries
- **Scope:** Robustness of birthplace favoritism in the modern, high-frequency era
- **Finding:** Favoritism starts **immediately on taking office** and persists after leaving it, with a further premium during COVID-19 in weaker democracies

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Data Sources: Outcome, Geography & Leaders

SOURCE	COVERAGE	USED FOR
OUTCOME VARIABLE		
01  DMSP-OLS Nighttime Lights 30 arc-sec (~1 km) · WGS84 (EPSG:4326) · geographic, unprojected (Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines, 2013)	Annual, 1992-2014 · 0-63 DN scale · sub-km pixels, all countries	Dependent variable: $\log(\text{light} + 0.01)$, average pixel intensity per region-year. A high concentration of observations sits at the lower bound, hence the added constant.
POLITICAL LEADERS		
02  Archigos, extended with birthplaces (Goemans et al., 2009)	Effective leader per country-year to 2004, hand-extended by HR to 2010 . Base universe: 919 leader-spells, 172 countries .	Defines who was leading which country, and when : the base list rows 3 and 4 attach a birthplace onto. Constructs Experience and TotalTenure directly; contributes no birthplace itself.
03  ↓ PLAD Birthplace Coordinates (Bomprezzi et al., 2024)	Matches 799 / 919 leader-spells (86.9%). Zero coverage for 25 countries .	Primary birthplace source for our own panel: coordinates point-in-polygon joined to GADM ADM2. Authoritative wherever it has an entry.
04  ↓ Wikidata/Wikipedia Scrape (Wikidata, 2026)	108 missing leaders queried, 42 resolved, 33 confirmed in-country.	Gap-fill only where PLAD has zero coverage. Combined coverage: $799 + 33 = 832 / 919 = 90.5\%$.
GEOGRAPHY		
05  Subnational Boundaries & Settlement Points WE USE GADM 3.6 Vector polygons · WGS84 (EPSG:4326) · geographic, unprojected (CIESIN, Columbia University, 2005) → (GADM, 2018)	Second-level administrative regions · 126 countries, 38,427 regions . Excludes countries under 500,000 people and any territory entirely above 65°N .	Defines HR's regional units of observation; settlement-point shapefiles geolocate leader birthplaces to a specific region. This CIESIN product was never publicly released as underlying data (SEDAC serves only PDF/PNG map previews), so our own geometry substitutes GADM 3.6 .
POPULATION		
06  Gridded Population of the World, v3 & v4 GPWv3 2.5 arc-min · GPWv4 5 arc-min · WGS84 (EPSG:4326), unprojected (CIESIN, Columbia University and CIAT, 2005) · (CIESIN, Columbia University, 2018)	Region-year · GPWv3 anchors 1990/1995, GPWv4 rev.11 anchors 2000/2005/2010, bridged and interpolated to an annual panel	Builds lnpop (log regional population in thousands) and the per-capita light outcome Lightpc . HR's Appendix A sources Population_ict from CIESIN "every fifth year from 1990," which is GPWv3; GPWv4 (2018) did not exist yet at HR's writing, so it extends the same CIESIN series past 2010. The two products' 2000 overlap year is used to compute a per-country bridge factor , scaling GPWv4 onto GPWv3's level before interpolating.

Data Sources: Country-Level Covariates

SOURCE	COVERAGE	USED FOR
INSTITUTIONS & HUMAN CAPITAL		
07  Political Regime Characteristics (Marshall et al., 2010)	Country-year, 1800-2010 · rescaled Polity2 score, 0 to 1	Measures executive constraints and openness of political competition; interacted with Leader in Table V .
08  Educational Attainment Dataset (Barro & Lee, 2013)	Country-year · 5-year intervals, 1950-2010	Average years of schooling attained; a second institutional-quality proxy interacted with Leader .
ECONOMIC & CULTURAL COVARIATES		
09  Penn World Table, GDP per Capita (Heston et al., 2012)	Country-year, U.S. dollars	Log GDP per capita , the baseline economic control for national income.
10  Linguistic Fractionalization (Alesina et al., 2003)	Country level, time-invariant	Probability two random citizens speak different languages; interacted with Leader as a cultural-cohesion proxy.
11  Family Ties (Alesina & Giuliano, 2014)	Country level, time-invariant	Strength of family bonds from World Values Survey responses; interacted with Leader alongside fractionalization.
EXTERNAL RESOURCE FLOWS		
12  Net Official Development Assistance (OECD, 2013)	Country-year, current U.S. dollars	Log Official Development Assistance per capita ; interacted with Leader in Table VII to test whether aid inflows fuel favoritism.
13  Adjusted Net Savings, Oil Rents (World Bank, 2013)	Country-year, current U.S. dollars per capita	Log oil rents per capita ; interacted with Leader alongside aid in the same heterogeneity test.
ALTERNATIVE OUTCOME (ROBUSTNESS)		
14  Subnational Regional GDP Data WE USE G-ECON 4.0 1 degree grid (~111 km) · WGS84 (EPSG:4326), unprojected (Gennaioli et al., 2014) → (Nordhaus et al., 2006)	1-degree global grid · benchmark years 1990/95/2000/05, 190 countries , aggregated to ADM2 by point-in-polygon join	Regional GDP per capita for Column (8). HR's own source, Gennaioli et al. (2014), is not publicly available . Reproduces the same direction and significance (0.013 vs. HR's published 0.021).

Data Sources: Descriptive Statistics

VARIABLES	Hodler & Raschky (2014) 126 countries · 1992-2009					Our Panel, HR-Exact Sample same 126 countries · 1992-2009				
	OBS.	MEAN	SD	MIN	MAX	OBS.	MEAN	SD	MIN	MAX
Light _{ict}	690,495	0.050	2.482	-4.605	4.143	655,659	0.098	2.456	-4.605	4.143
Leader _{ict-1}	690,495	0.004	0.060	0.000	1.000	613,433	0.004	0.060	0.000	1.000
Polity _{ct-1}	684,213	0.789	0.257	0.000	1.000	612,746	0.789	0.258	0.000	1.000
	(2,205)	(0.670)	(0.312)			(1,922)	(0.680)	(0.307)		
Schooling _{ct-1}	648,240	8.168	2.783	0.925	13.022	586,029	8.124	2.778	0.930	13.052
	(1,922)	(6.968)	(2.920)			(1,700)	(7.081)	(3.015)		
NationalGDP _{ct-1}	683,669	8.868	1.147	5.080	10.855	615,481	8.829	1.178	5.081	10.855
	(2,200)	(8.212)	(1.347)			(1,946)	(8.245)	(1.356)		
Language _c	679,119	0.318	0.281	0.002	0.923	647,822	0.313	0.286	0.002	0.923
	(2,161)	(0.438)	(0.298)			(2,022)	(0.442)	(0.297)		
FamilyTies _c	551,004	0.096	0.311	-0.851	0.606	521,800	0.059	0.365	-0.912	0.572
	(1,112)	(0.041)	(0.368)			(1,021)	(-0.015)	(0.391)		
Aid _{ct-1}	690,495	6.191	4.743	-10.632	13.712	405,898	7.891	3.284	-10.597	13.559
	(2,251)	(7.794)	(4.746)			(1,476)	(9.262)	(2.651)		
Oil _{ct-1}	645,396	9.357	4.116	0.000	16.396	609,906	8.945	4.038	0.000	16.126
	(1,820)	(6.590)	(5.262)			(1,900)	(5.954)	(5.146)		

Notes. Descriptive statistics at the country-year level in parentheses.

Our panel restricted to HR's own **126-country list** and their exact **1992-2009** window, ADM2 regions entirely above **65°N** dropped, matching their stated exclusion criteria exactly, with the same variables and **t-1** lag structure. Remaining gaps in Obs. reflect **real source-level coverage differences** (e.g. GPWv3/v4 vs. HR's own CIESIN vintage), not sample-restriction bugs.

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Empirical Framework



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$$\text{Light}_{ict} = \alpha_i + \beta_{ct} + \gamma \text{Leader}_{ict-1} + \varepsilon_{ict}$$

THE VARIABLES

- Light_{ict} is log nighttime light intensity($\log(\text{light} + 0.01)$) for region i , country c , year t .
- Leader_{ict-1} is 1 if region i was the effective leader's birth region in $t - 1$, 0 otherwise.

FIXED EFFECTS, α_i AND β_{ct}

- α_i absorbs everything **time-invariant** about a region.
- β_{ct} absorbs every **national shock common to a country-year**.
- What survives is variation **within region, within country, within year**.

COEFFICIENT OF INTEREST, γ

- HR expect and find $\gamma > 0$: a **positive, significant** γ is read as evidence of regional favoritism.
- Standard errors are **clustered by leader**, lagged one period for consistency with Leader_{ict-1} itself.

Main Results and Robustness Tests (Table II)

	(1)		(2)		(3)		(4)		(5)		(6)		(7)		(8)	
	Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}		Light0 _{ict}		Lightpc _{ict}		RegionalGDP _{ict}	
	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.
Leader _{ict-1}	0.038*** (0.014)	0.032** (0.015)					0.019*** (0.010)	0.026*** (0.010)	0.061*** (0.010)	0.050*** (0.008)	0.029*** (0.013)	0.008 (0.012)	0.062*** (0.024)	0.025** (0.011)	0.021*** (0.006)	0.013** (0.006)
Leader _{ict}			0.039*** (0.015)	0.029* (0.016)												
Leader _{ict-2}					0.041*** (0.013)	0.040*** (0.015)										
Light _{ict-1}							0.400*** (0.023)	0.408*** (0.025)	0.962*** (0.004)	0.964*** (0.004)						
Pop _{ict}													0.958*** (0.066)	-0.714*** (0.067)	0.201*** (0.049)	-0.007* (0.004)
Number of regions	38,427	38,474	38,427	38,474	38,427	38,474	38,427	38,474	38,427	38,474	36,591	37,179	37,475	38,474	1,207	6,531
Observations	690,495	654,254	690,495	655,234	689,870	578,100	652,362	616,760	652,362	616,760	619,594	593,814	673,382	654,254	14,995	113,110
Within R ²	0.319	0.314	0.319	0.315	0.318	0.239	0.412	0.398	0.964	0.969	0.393	0.350	0.197	0.418	0.653	0.803
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions (except for Column (5), which is standard OLS) using annual data for subnational regions between 1992 and 2009. **Light_{ict}** is the log of average nighttime light intensity plus 0.01. **Light0_{ict}** is the log of average nighttime light intensity (without adding a constant). **Lightpc_{ict}** is the log of nighttime light intensity per capita plus 0.01. **RegionalGDP_{ict}** is the log of regional GDP per capita. **Leader_{ict}** is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. **Pop_{ict}** is the log of regional population. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

Sample restricted to HR's exact **126-country, 1992-2009** window. Column (8) uses G-Econ 4.0, our substitute for HR's own Gennaioli et al. (2014) source, which was never publicly released. All eight **Leader** coefficients match HR's sign; Column (6) **loses significance** in our panel (0.008, not significant, vs. HR's 0.029***). **Pop**'s sign flips in Columns (7)-(8), a genuine population-source difference (GPWv4 vs. HR's CIESIN vintage), not a bug.

Empirical Framework: Geographic Extent (Table IV)

														
	(1)		(2)		(3)		(4)		(5)		(6)		(7)	
	Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}		Light _{ict}	
	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.	ORIG.	REPL.
Leader_{ict-1}	0.049** (0.024)	0.024 (0.020)	0.026** (0.010)	0.017 (0.011)	0.023** (0.012)	0.010 (0.012)	0.043*** (0.010)	0.036*** (0.012)	0.028** (0.011)	0.019* (0.011)	0.020** (0.009)	-0.001 (0.009)	0.005 (0.009)	-0.004 (0.008)
Units of observation	5 km circles		SN1 regions		SN1, excl. birth-SN2		50 km grid		100 km grid		200 km grid		400 km grid	
Number of regions	520		2,242		2,220		64,204		17,242		6,101		2,110	
Observations	9,134		40,157		39,761		1,048,370		280,210		101,655		35,889	
Within R ²	0.520		0.602		0.603		0.210		0.298		0.362		0.449	
Region FE	Yes		Yes		Yes		Yes		Yes		Yes		Yes	
Country-year FE	Yes		Yes		Yes		Yes		Yes		Yes		Yes	

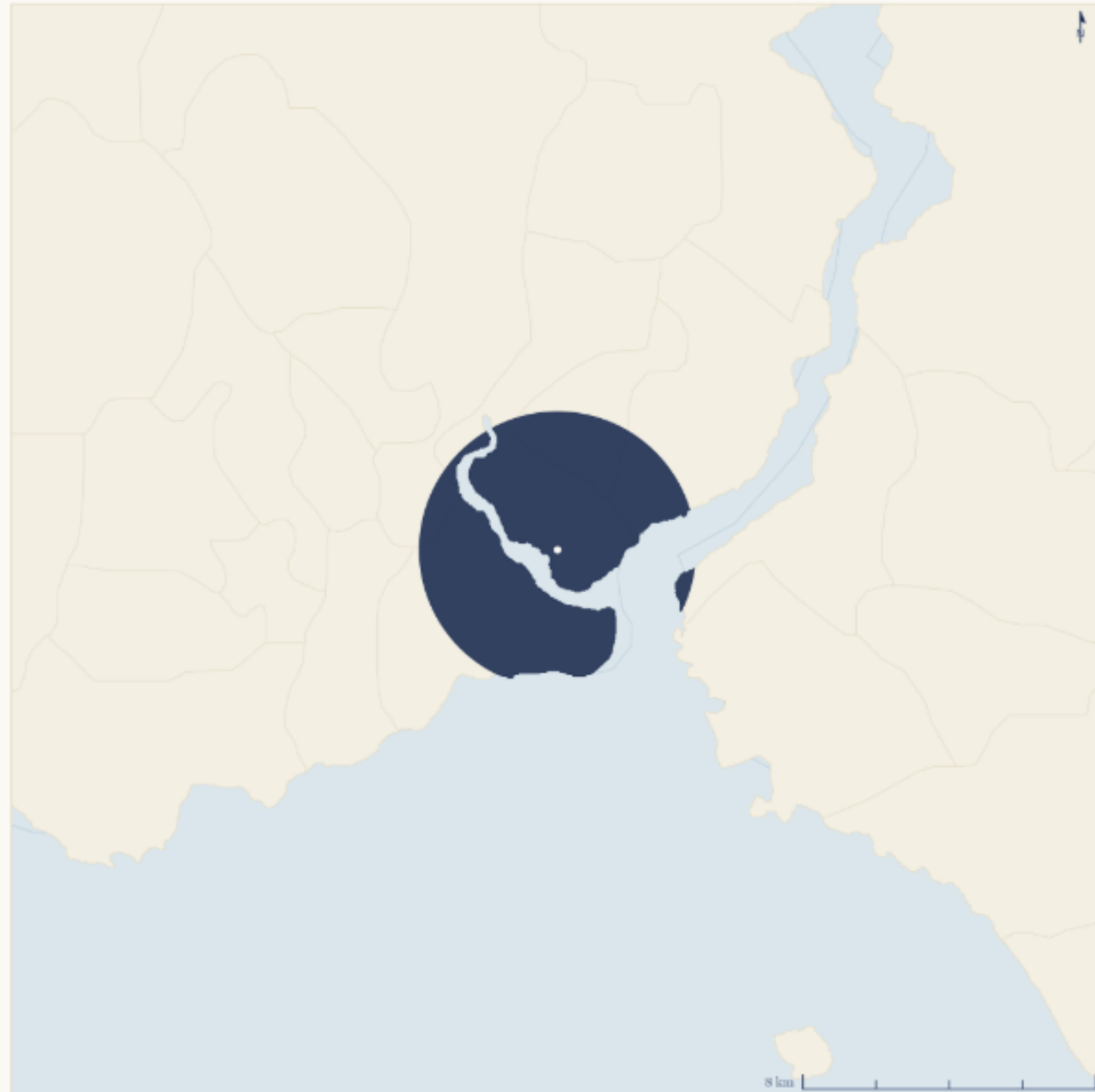
Notes. Fixed effect regressions using annual data for subnational regions between 1992 and 2009. The units of observation differ across columns and are indicated in their own row (see text for details). **Light_{ict}** is the log of average nighttime light intensity plus 0.01. **Leader_{ict}** is a dummy variable equal to 1 if region *i* is the birth region of the political leader in country *c* in year *t*, and 0 otherwise. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

Sample restricted to HR's exact **126-country, 1992-2009** window. Each column changes only the **size and shape of the spatial unit**, not the specification: from a tight 5 km circle around the leader's birthplace, to the full home region, to the home region with the birth district cut out, to a world grid at four coarsening resolutions. The **Leader coefficient shrinks toward zero as the unit gets larger** in both HR's numbers and ours, the same pattern the paper reports: favoritism shows up most clearly close to the leader's birthplace and fades out as the unit of observation grows past it.



5 km circles

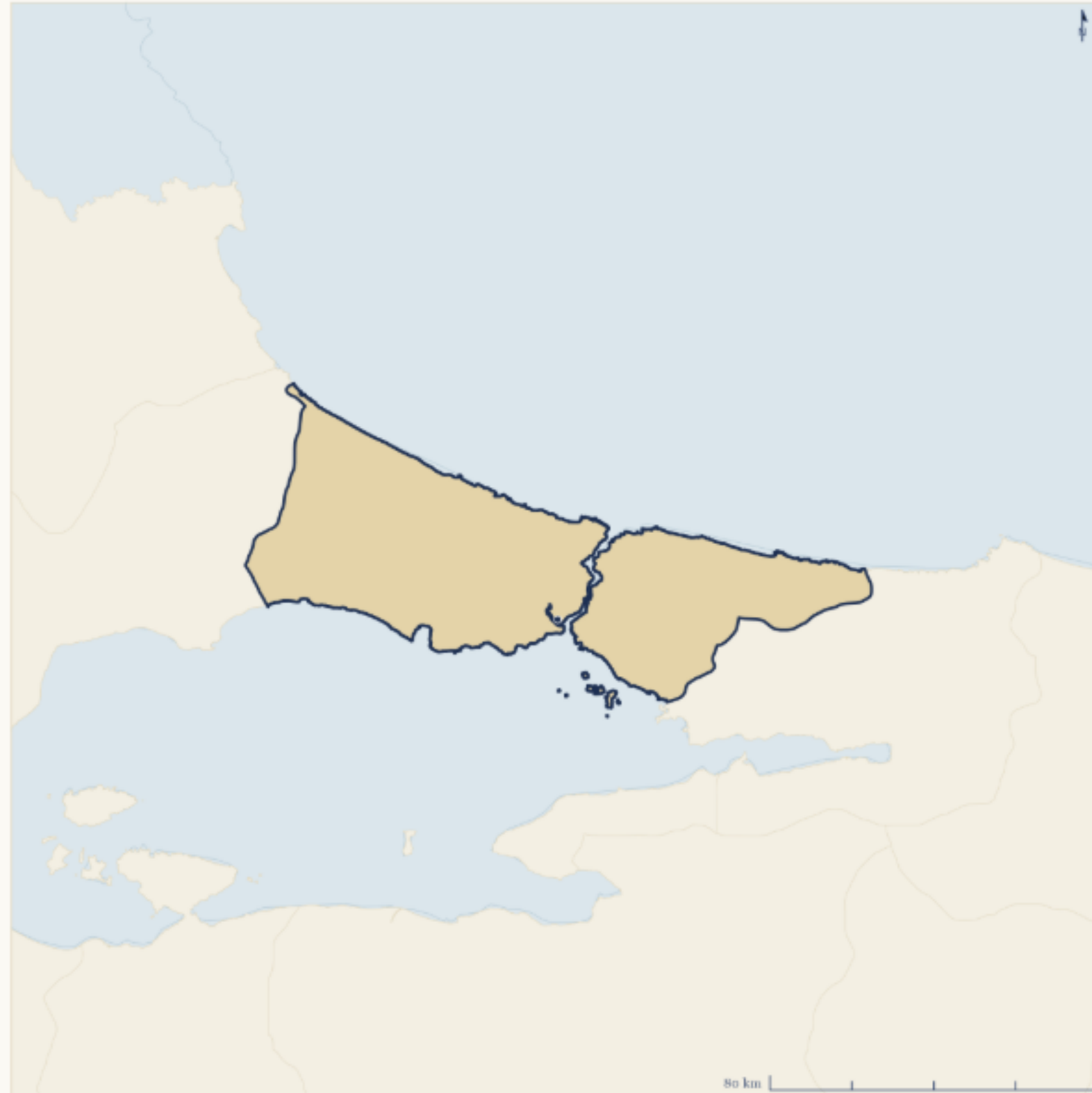
Geodesic buffer around the birth coordinate, clipped at the coastline · HR Col (1)





SN1 regions

Istanbul, 1 of Turkey's 81 first-level administrative regions · HR Col (2)





SN1, excl. birth-SN2

Beyoğlu (Erdoğan's birth ADM2) geometrically cut out · HR Col (3)





50 km grid

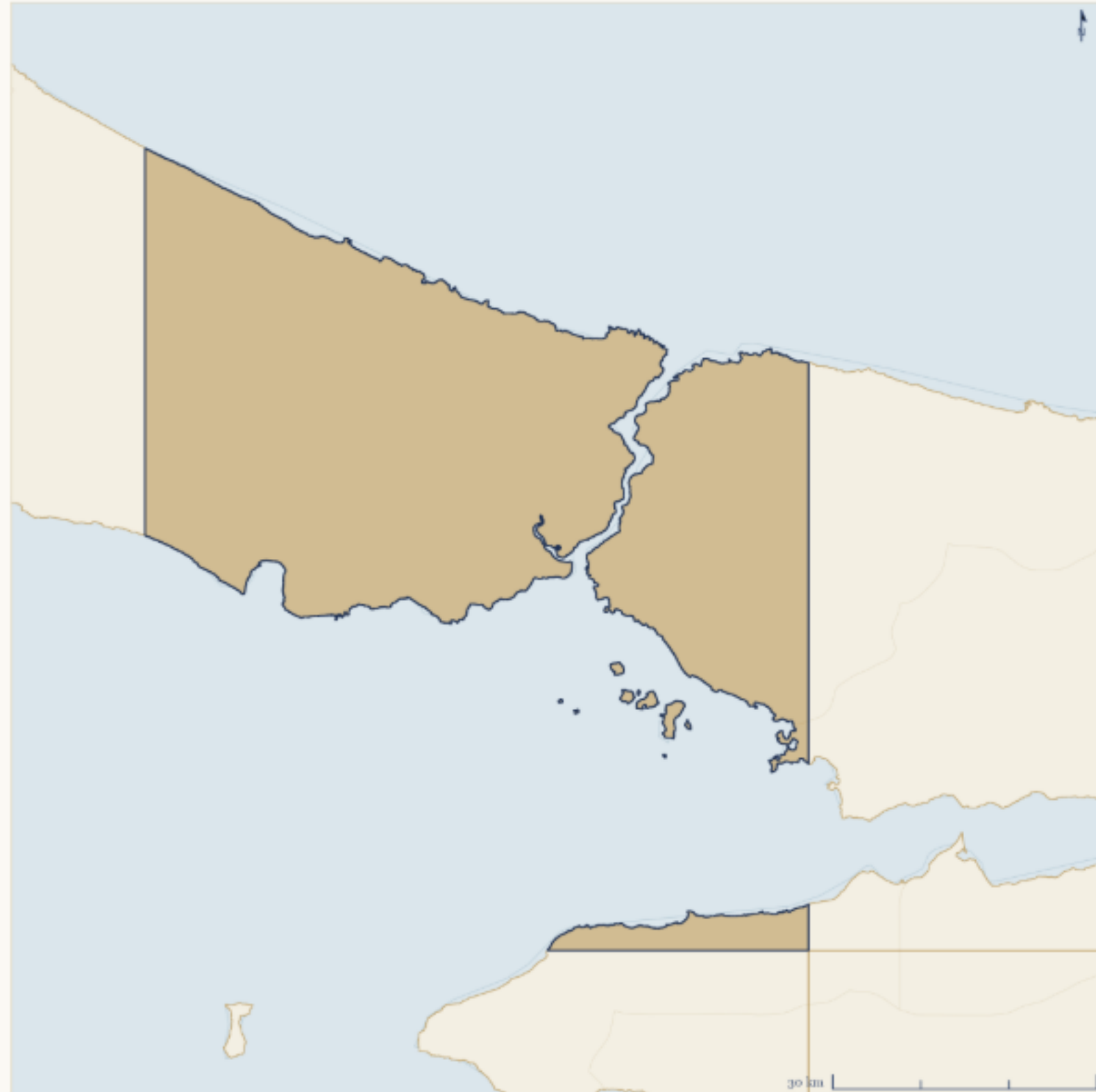
Equal-area cell containing the birth point, clipped at coastlines/borders · HR Col (4)





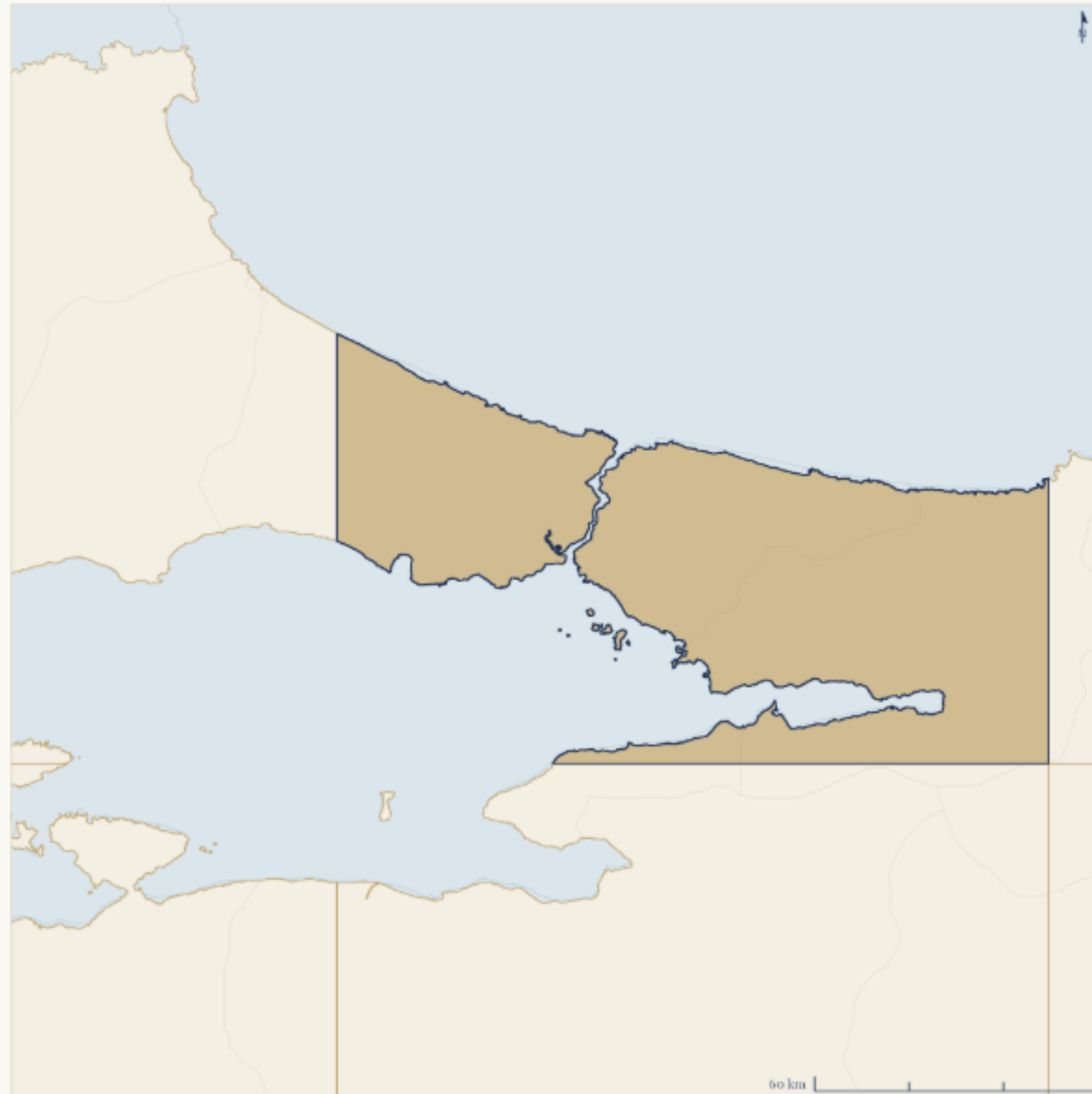
100 km grid

Equal-area cell containing the birth point, clipped at coastlines/borders · HR Col (5)



200 km grid

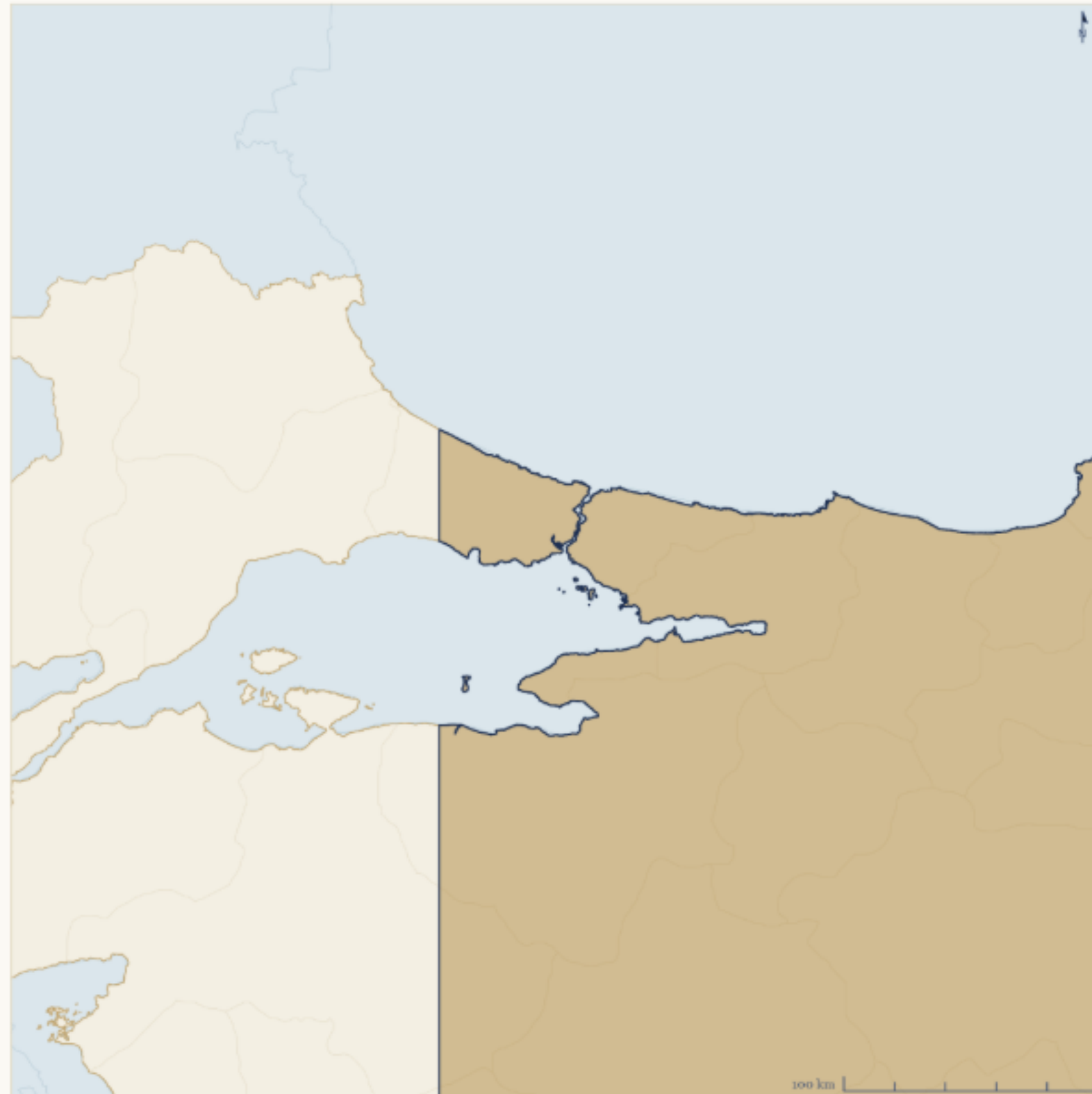
Equal-area cell containing the birth point, clipped at coastlines/borders · HR Col (6)





400 km grid

Equal-area cell containing the birth point, clipped at coastlines/borders · HR Col (7)



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Extension: Beyond HR's Original Window

1992–2023

TIME PERIOD

Harmonized **DMSP¹/VIIRS²** panel, Li et al. ([Li et al., 2020](#)): 30 arc-sec (~1 km) WGS84 GeoTIFF, 0-63 DN scale, nearly double HR's own 1992-2009 window. VIIRS (2012 onward) is the higher-resolution successor sensor to DMSP, natively 15 arc-sec (~500 m), with no saturation over bright urban cores.

¹ EOG DMSP (Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines, 2013) · ² EOG VIIRS (Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines, 2012)

GHS-POP

POPULATION · 1990–2020

Own pipeline ([Schiavina et al., 2023](#)): 3 arc-sec (~100 m) WGS84 GeoTIFF rasters, zonal-summed to ADM2, linearly interpolated to annual values.

PLAD + Wikidata

LEADER DATA

Same source as the main replication ([Bomprezzi et al., 2024](#); [Wikidata, 2026](#)), extended to the full window.

Extension: Results Over the Full 1992-2023 Window

	(1) Light _{ict}	(2) Light _{ict}	(3) Light _{ict}	(4) Light _{ict}	(5) Light _{ict}	(6) Light _{0ict}	(7) Light _{pcict}
Leader _{ict-1}	0.048*** (0.017)			0.025** (0.010)	0.074*** (0.016)	0.035** (0.016)	0.026*** (0.009)
Leader _{ict}		0.043** (0.017)					
Leader _{ict-2}			0.046** (0.018)				
Light _{ict-1}				0.556*** (0.026)	0.943*** (0.009)		
Pop _{ict}							-0.534*** (0.058)
Number of regions	45,463	45,463	45,463	45,463	45,463	45,133	45,386
Observations	1,301,050	1,360,509	1,274,892	1,260,294	1,260,314	1,186,109	1,298,598
Within R ²	0.516	0.461	0.435	0.634	0.945	0.604	0.472
Region FE	Yes	Yes	Yes	Yes	No	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Fixed effect regressions on the harmonized DMSP/VIIRS panel, 1992-2023 (except Column (5), which is standard OLS). Same seven specifications as the main replication’s Table II, Columns (1)-(7); Column (8), which relies on G-Econ regional GDP, is not extended. Population is our own independently-built GHS-POP source. Standard errors clustered at the country level. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

The core result **holds up over the extended 1992-2023 window**, using entirely our own extension-era data pipeline: the Leader coefficient is positive and significant across every specification, contemporaneous (Col 2), two-year lag (Col 3), and the dynamic panel (Col 4). Column (6), the extensive margin, is the one column that **failed to replicate** on HR’s own 1992-2009 window; here it **recovers to significance**, consistent with a DMSP sensor floor rather than a fragile finding. Column (7) per capita is the most stable result in the table: 0.026, almost unchanged from 0.025 on the shorter window, despite an entirely different population source and estimation period.

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Conclusion



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The replication holds

- HR 2014's central finding, that political leaders' birth regions receive brighter nighttime lights while they are in power, **reproduces cleanly** in our own independently-assembled data: same sign, same order of magnitude, across every geometric definition of "region" we tried, from a 5 km circle to a 400 km grid cell
- The few places our numbers diverge from HR's own (Table II Col 6, the Pop sign flip in Cols 7-8) trace to **documented, real source-vintage differences**, not construction bugs

The extension confirms it, not just once

- The same Leader effect **survives an independently-rebuilt population pipeline** (GHS-POP) and a **32-year window** nearly double HR's original span, with the effect if anything slightly stronger
- Both the original replication and the extension land on the **same qualitative story**: birth-region favoritism in nighttime light is real, robust to spatial-unit choice, and not an artifact of any one data vintage or time window

References

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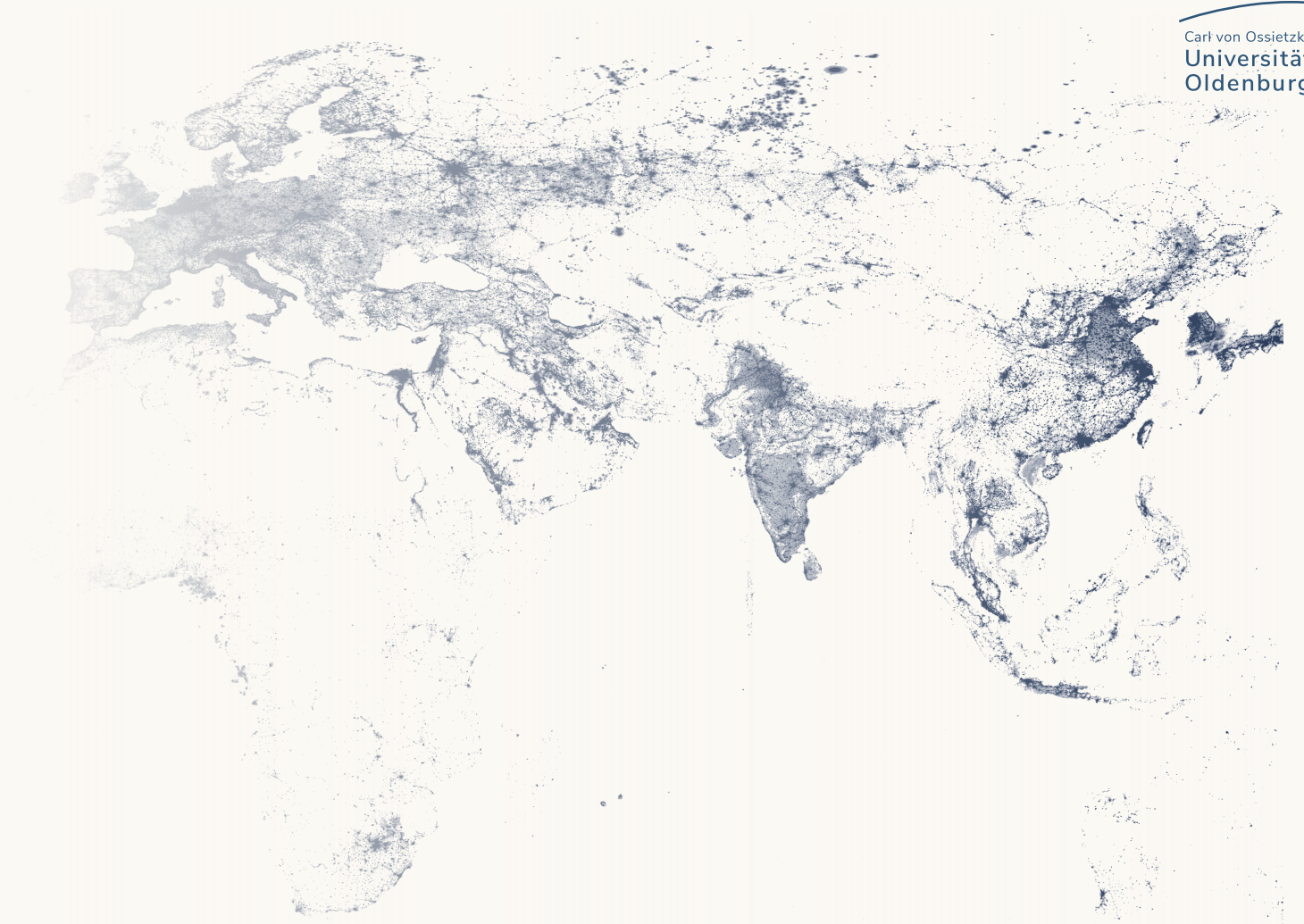
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Thank You

Questions?



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